triple integral

evaluate the iterated integral $\int_0^{\pi/2} \int_0^{2x} \int_0^{x+z} \cos(x-2y+z)\, dy\, dz\, dx$

$$\int_0^{x+z} \cos(x-2y+z)\, dy$$

$u = x-2y+z$

$du = -2dy$

$\frac{du}{-2} = dy$

$$\int_0^{x+z} \cos(u) \cdot \frac{du}{-2}$$

$$-\frac{1}{2}\left[\sin(x-2y+z)\right]_0^{x+z} \Rightarrow -\frac{1}{2}\left[\sin(x-2x+3z) - \sin(x+z)\right]$$

$$-\frac{1}{2}\int_0^{2x}\left[\sin(-x+3z) - \sin(x+z)\right] dz \Rightarrow -\frac{1}{2}\int_0^{2x} \sin(-x+3z)\, dz - \frac{1}{2}\int_0^{2x} \sin(x+z)\, dz$$

$u = -x+3z$

$du = 3dz$

$\frac{du}{3} = dz$

$u = x+z$

$du = dz$

$$-\frac{1}{6}\int_0^{2x} \sin(u)\, du - \frac{1}{2}\int_0^{2x} \sin(u)\, du$$

$$-\frac{1}{6}\left[-\cos(-x+3z)\right]_0^{2x} - \frac{1}{2}\left[-\cos(x+z)\right]_0^{2x}$$

$$-\frac{1}{6}\left[-\cos(-x+6x) + \cos(-x)\right] - \frac{1}{2}\left[-\cos(3x) - 1\right]$$

$$\int_0^{\pi/2} \frac{\cos(5x)}{6} - \frac{\cos(-x)}{6} + \frac{\cos(3x)}{2} + \frac{1}{2}\, dx$$

$$\int_0^{\pi/2} \frac{\cos(5x)}{6}\,dx - \int_0^{\pi/2} \frac{\cos(-x)}{6}\,dx + \int_0^{\pi/2} \frac{\cos(3x)}{2}\,dx + \int_0^{\pi/2} \frac{1}{2}\,dx$$

apply u- sub

$$\frac{1}{30}\int_0^{\pi/2} \cos(u)\,du + \frac{1}{6}\int_0^{\pi/2} \cos(u)\,du + \frac{1}{6}\int_0^{\pi/2} \cos(u)\,du + \int_0^{\pi/2} \frac{1}{2}\,dx$$

$$\frac{1}{30}\Big[\sin(5x)\Big]_0^{\pi/2} + \frac{1}{6}\Big[\sin(-x)\Big]_0^{\pi/2} + \frac{1}{6}\Big[\sin(3x)\Big]_0^{\pi/2} + \Big[\frac{1}{2}x\Big]_0^{\pi/2}$$

$$\frac{1}{30}\sin\left(\frac{5\pi}{2}\right) + \frac{1}{6}\sin\left(-\frac{\pi}{2}\right) + \frac{1}{6}\sin\left(\frac{3\pi}{2}\right) + \frac{\pi}{4}$$

$$\frac{1}{30} - \frac{1}{6} - \frac{1}{6} + \frac{\pi}{4} = \boxed{\frac{\pi}{4} - \frac{3}{10}}$$

evaluate the triple integral where $E = \{(x,y,z) \mid 0 \le y \le 1,\ y \le x \le 1,\ 0 \le z \le xy\}$

one thing we already know is the one with numbers must be the outer one while you now need to figure out what order the inner ones go in

$$\int_0^1 \qquad\qquad dy$$

We place the one with only numbers outside and it goes by dy now here comes the trick that will help us figure out exactly which one comes next, we already know that this goes by dy so the next integral needs to have limits that have only a y variable

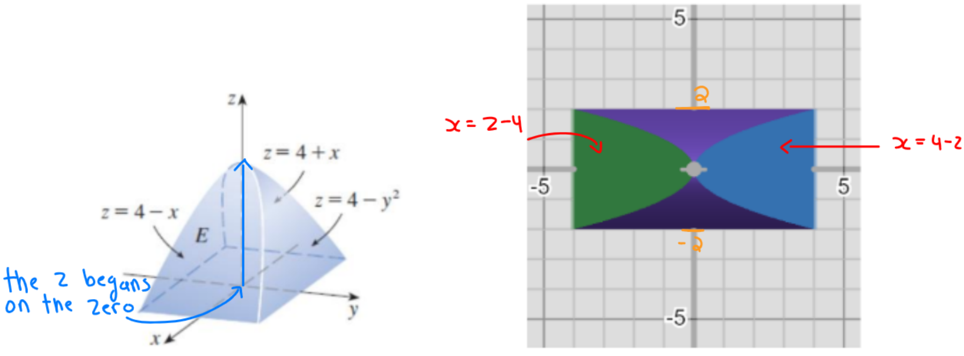
$$\int_0^1 \int_y^1 \qquad dx\,dy$$

So we Found the Next integral and notice it goes by dx So For the Final integral (there is only one left) you want to Find a integral that has x in its limits

$$\int_0^1 \int_y^1 \int_0^{xy} dz\, dx\, dy$$

Now we Finished with the last integral notice in the limits it has xy this can happen and will happen but you should be fine Just Follow the logic we applied here

3. Evaluate the integral $\iiint_E 12\, dV$ where the solid E id is given by the picture below.



$z-4 \leq x \leq 4-z$

$-2 \leq y \leq 2$

$0 \leq z \leq 4-y^2$

- We know this is bounded by y because if we look at the left graph y is Flat on ground the only movment along y show that its bound at -2 to 2 which we see on the right graph

We already know the $-2 \leq y \leq 2$ will be on the outer integral because its a number
We will apply the same reasoning and logic we did on the last problem

$$\int_{-2}^{2} \qquad dy$$

now find the integral limits that have the variable y in them

$$\int_{-2}^{2} \int_{0}^{4-y^2} \qquad dz\, dy$$

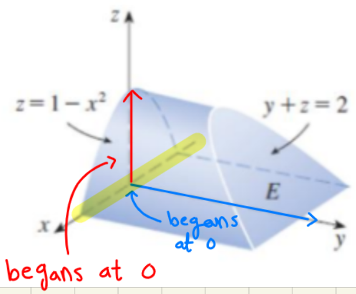
We got the second integral with its limits notice this goes in the dz direction
now find the integral limits that have z in them

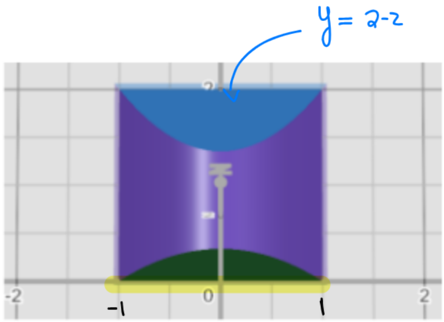
$$\int_{-2}^{2} \int_{0}^{4-y^2} \int_{z-4}^{4-z} \qquad dx\, dz\, dy$$

ok we know have found the final inner integral limit this integral was only set up but not solved, solving this integral is the same as any triple integral

$$\int_{-2}^{2} \int_{0}^{4-y^2} \int_{z-4}^{4-z} 12 \; dx\, dz\, dy$$

4. Evaluate the integral $\iiint_E 3x\, dV$ where the solid E id is given by the picture below.





this is bounded by the x-axis

$-1 \le x \le 1$

$0 \le y \le 2-z$

$0 \le z \le 1-x^2$

- the outer integral need to be from −1 to 1

$$\int_{-1}^{1} \quad dx$$

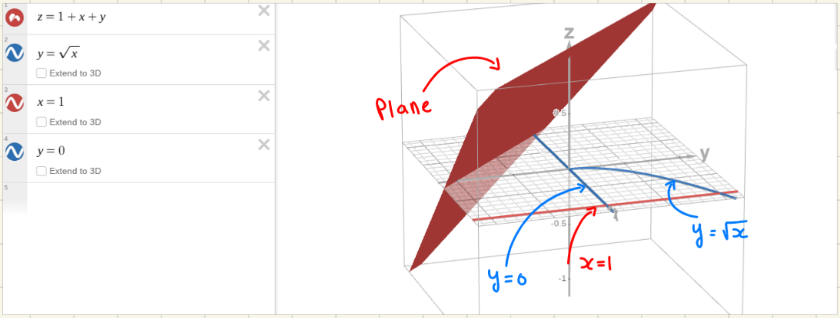
the outer integral has dx so the Next integral limits needs to have the variable x in it

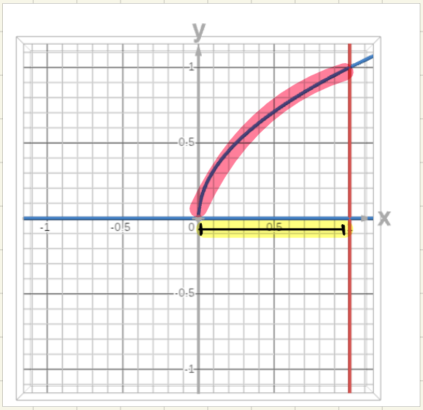
$$\int_{-1}^{1}\int_{0}^{1-x^2} \quad dz\, dx$$

the second inner integral has dz so the Final integral needs to have a variable z inside the limit of integral

$$\int_{-1}^{1}\int_{0}^{1-x^2}\int_{0}^{2-z} 3x\, dy\, dz\, dx$$

$\iiint_E 6xy\,dV$, where E lies under the Plane $z=1+x+y$ and above the region in the xy-Plane bounded by the curves $y=\sqrt{x}$, $y=0$, and $x=1$





$0 \le x \le 1$

$0 \le y \le \sqrt{x}$

$0 \le z \le 1+x+y$

$$\int_0^1 \int_0^{\sqrt{x}} \int_0^{1+x+y} 6xy\,dz\,dy\,dx$$

to Put the integral togther we applied the same ideas as before We already know the 0 to 1 need to be on the outer integral and it goes by dx so the Second integral limits need to have x in them so we got 0 to $\sqrt{x}$ for the Second integral and it goes by dy So the Next integral Needs to have y in it So We got 0 to $1+x+y$ even though it has x in it does not Matter that will happen From time to time